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MAJOR PROJECT DESIGN & ARCHITECTURE PRESENTATION

Robo-Root: Plant-Root-Inspired Soft Growing Robot

Pneumatic Tip Eversion, Base-Mounted Four-Tendon Steering & Vision-Assisted Autonomous Navigation

PROJECT MEMBER
Mihal Palakkal
Reg No: 20223071

PROJECT MEMBER
Rose Maria
Reg No: 20223085

PROJECT MEMBER
Ruvais P
Reg No: 20223087

PROJECT MEMBER
Mohamed Jaseeb P J
Reg No: 20223120

ACADEMIC ADVISORY & PROJECT SUPERVISION

Ms. Amrutha S Nair
Project Guide
Assistant Professor, Dept. of CSE

Dr. Sheena S
Project Coordinator
Associate Professor, Dept. of CSE

Mr. Damodaran V
Head of Division
Division of Computer Science & Engg

Bio-Inspired Soft Robotics: Why Tip-Growing Systems?

Overcoming the severe physical constraints of rigid locomotion in complex and confined subterranean environments.

LIMITATIONS OF CONVENTIONAL RIGID ROBOTS

- **Whole-Body Motion & Friction:** Wheeled, tracked, or legged robots must drag or propel their entire body mass forward through the terrain, creating immense frictional drag in narrow passages.
- **Wedging & Geometric Trapping:** Rigid structures cannot deform dynamically through tortuous or collapsing cavities, leading to entrapment in disaster rubble or tight pipes.
- **Subterranean Soil Resistance:** Penetrating soil or sand with rigid shafts incurs exponential penetration resistance that scales drastically with penetration depth.
- **High In-Tip Actuator Mass:** Locating heavy drive motors or articulating joints near the front compromises balance and severely limits miniaturization.

THE ROBO-ROOT ADVANTAGE: APICAL TIP EVERSION

✔️ **Apical Extension (Zero Wall-Sliding Friction):** Inspired by biological plant roots, material extends strictly at the tip through pneumatic eversion. The deployed body remains stationary relative to surrounding walls.

✔️ **Passage Through Unstructured Voids:** Seamlessly adapts to tortuous pathways, squeezing through apertures narrower than the unpressurized tube diameter.

✔️ **Internal Working Lumen Conduit:** The hollow central lumen provides a protected interior pathway for electrical wiring, sensor payloads, and air supply without environmental exposure.

TARGET APPLICATIONS:
Urban Search & Rescue in collapsed structures, industrial pipeline and duct inspection, underground sensor deployment, and non-destructive soil probing.

Core Engineering & Scientific Challenges

Balancing pneumatic growth dynamics, tendon-based mechanical steering, and real-time vision perception.

1. DIRECTIONAL STEERING CONTROL

• Uncontrolled Tip Eversion: Pressurized pneumatic tubes naturally extend along the path of least resistance or in a straight line.

• Planar & Spatial Bending: Imparting deliberate 2D/3D angular deflection requires active, coordinated steering authority without inducing structural buckling or tube kinking.

• Critical Need: A decoupled 4-tendon antagonistic steering mechanism operating at the tip.

2. TENDON ROUTING & CABLE PAYOUT

• Growth vs. Cable Tension: As the robot extends from centimeters to meters, steering cables must deploy continuously.

• Slack & Friction Management: Loose tendons cause steering deadband and backlash, while overtight tendons prevent eversion growth.

• Critical Need: Base-mounted motorized cable spools that dynamically release tendon slack synchronous with growth.

3. TIP PERCEPTION IN CONFINED SPACES

• Hostile Visual Environments: Confined boreholes and rubble exhibit complete darkness, dust, lens occlusions, and severe glare.

• Computational Bottlenecks: Real-time image processing cannot rely on heavy GPU hardware mounted on the fragile soft tip.

• Critical Need: Micro-camera feed streaming to a host ROS 2 machine running optimized YOLO inference.

4. MULTIMODAL SYSTEM SYNCHRONIZATION

• Actuator-Sensor Decoupling: High pneumatic pressure, servo pulse modulation, and image capture must be synchronized.

• Closed-Loop Control Latency: Steering corrections must occur at speeds compatible with pneumatic forward advancement.

• Critical Need: A modular ROS 2 architecture establishing deterministic communication across all subsystems.

Systematic Objectives & Technical Scope Boundaries

Defining the engineering milestones and practical boundaries of the laboratory prototype.

PRIMARY TECHNICAL OBJECTIVES

1. Pneumatic Tip Eversion: Design an airtight, flexible tubular mechanism capable of continuous forward growth driven by regulated pneumatic pressure.

2. Four-Tendon Steering: Construct a 4-tendon orthogonal arrangement around the flexible tube for active planar tip steering in the X-Y horizontal plane.

3. Base-Station Actuation: Mount four high-torque servo motors at the stationary base to minimize tip payload and enable active tendon spooling.

4. Vision Integration: Equip the advancing tip with a micro-camera for real-time forward environmental sensing and obstacle detection.

5. YOLO Perception Pipeline: Integrate pretrained YOLO models via OpenCV to detect objects and obstacles without needing massive prior training sets.

6. ROS 2 Control Architecture: Establish a modular ROS 2 framework linking image acquisition, path planning, kinematics, and ESP32 low-level firmware.

SCOPE BOUNDARIES & ASSUMPTIONS

- **Base-Mounted Servos:** Servos remain stationary at the base rather than moving with the tip, drastically reducing moving inertia but requiring robust cable payout management.
- **Pretrained Model Utilization:** Leverages established YOLO weights for standard object classes, focusing on the system integration pipeline rather than de novo neural network training.
- **Perception Outputs:** Visual detection yields 2D bounding boxes and class IDs; metric depth estimation is handled modularly through geometric or additional depth sensors.
- **Empirical Validation Focus:** Prototype designed for laboratory bench testing to evaluate growth rate, steering deflection angle, and deployment-retrieval cycles.

End-to-End System Workflow & ROS 2 Framework

Modular software nodes communicating across perception, path planning, and embedded actuation.

INTEGRATED SYSTEM WORKFLOW LOOP

1. Initialization: ROS 2 nodes, camera stream, ESP32 UART, & pneumatic manifold initialize.

2. Pneumatic Growth: Regulated air enters chamber; tubular membrane begins tip eversion.

3. Frame Acquisition: Tip camera captures visual frames ahead of advancing robot.

4. YOLO Inference: OpenCV resizes/letterboxes frames; YOLO outputs bounding boxes.

5. Path Planning: ROS 2 node determines obstacle-free trajectory setpoint.

6. Steering Kinematics: Controller converts target angle into 4 differential tendon lengths.

7. Servo Actuation: ESP32 generates PWM pulses for 4 base-mounted servo motors.

8. Cable Payout: Motorized drums release tendon slack synchronous with tube growth.

9. Pressure & Safety: Pressure transducer ensures line pressure stays within safe thresholds.

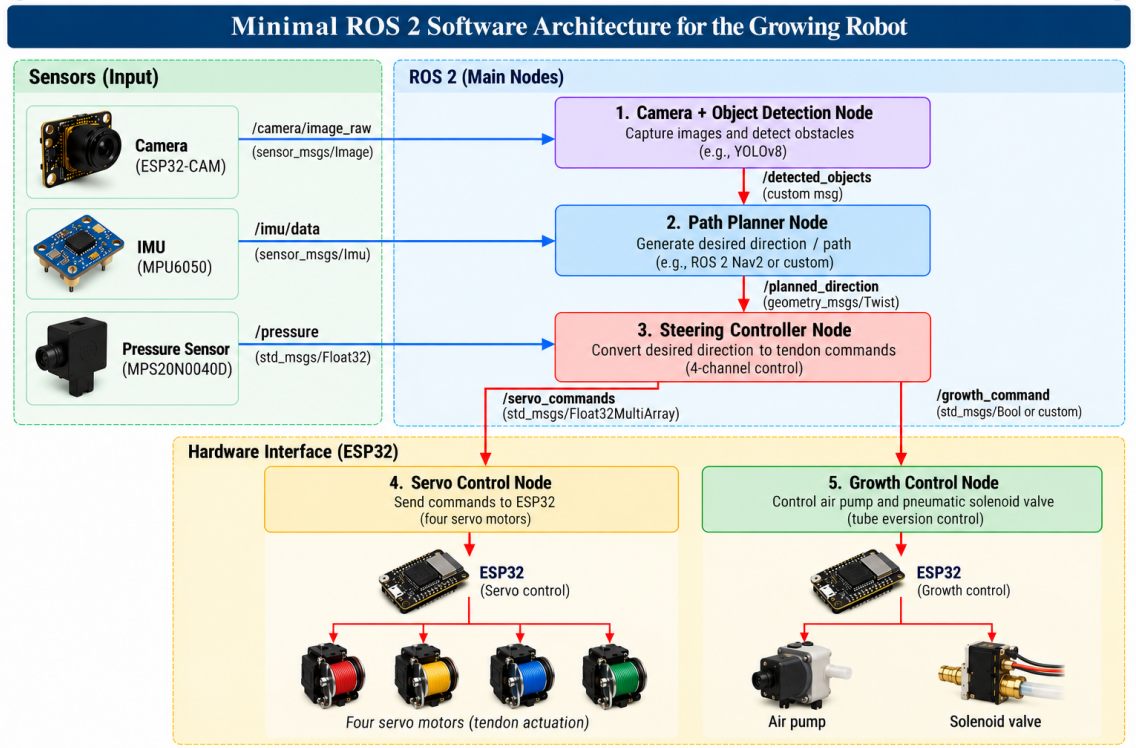


Figure 5.1: ROS 2 Software Architecture & Hardware Interface

Mechanical Design & Tip-Eversion Mechanism

Pneumatic inversion dynamics, base-station structural layout, and internal lumen routing.

MECHANICAL DESIGN PRINCIPLES

- **Tip Eversion Dynamics:** The robot body consists of a flexible, airtight membrane folded inwards upon itself. Pneumatic pressure applied internally forces new material to unfold exclusively at the tip, pushing the robot forward without sliding friction.
- **Base Station Assembly:** Houses the primary structural weight: the air supply connection, pressure regulator, four servo actuators, cable payout drums, and control electronics. Isolates delicate parts from hazardous environments.
- **Growing Tip Head Assembly:** A lightweight, rigid circular ring mounted at the inversion point. Acts as the anchor for the 4 steering tendons, mounts the micro-camera module, and ensures circular aperture maintenance.
- **Internal Working Lumen:** The hollow central core forms a protected conduit that carries tendon cables, camera wiring, and pneumatic hoses without exposing them to exterior dirt, sand, or debris.

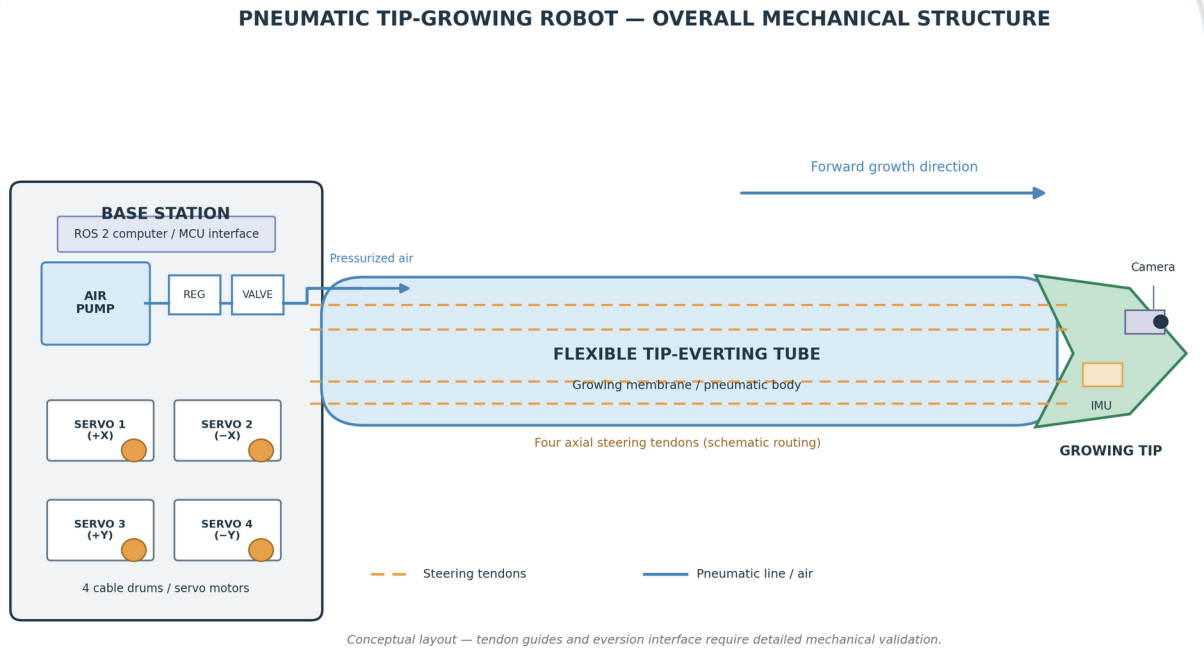


Figure 6.1: Overall Mechanical Structure of the Pneumatic Tip-Growing Robot

Four-Tendon Steering & Base-Mounted Servo Mechanism

Decoupling actuation weight from the tip through base-mounted antagonistic cable spools.

DIRECTIONAL STEERING & ACTUATION

- **Four-Tendon Orthogonal Arrangement:** Four axial cables are routed circumferentially at 90° intervals (North, East, South, West). Antagonistic tensioning allows omnidirectional 2D bending in the X-Y plane.
- **Differential Bending Moments:** Pulling tendon X+ while releasing tendon X- creates a bending curvature in the +X direction without altering overall growth speed or internal pressure.
- **Base-Mounted Actuator Spools:** Mounting the four high-torque servo motors at the stationary base keeps tip inertia at zero, preventing front-heavy tipping and allowing high steering agility.
- **Dynamic Cable Payout Mechanics:** Each servo drives a precision cable drum with guide channels. As the robot grows, tendons pay out in lockstep with the advancing membrane, maintaining consistent tension and avoiding tangling.

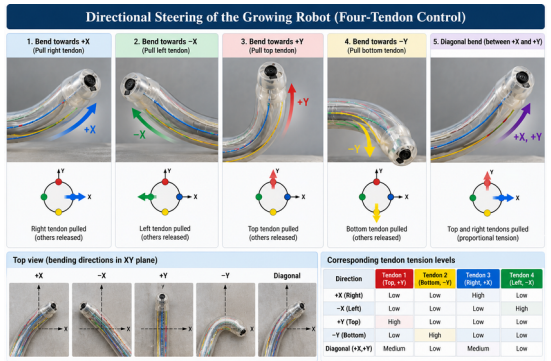


Figure 6.2: Directional Steering via Four-Tendon Control

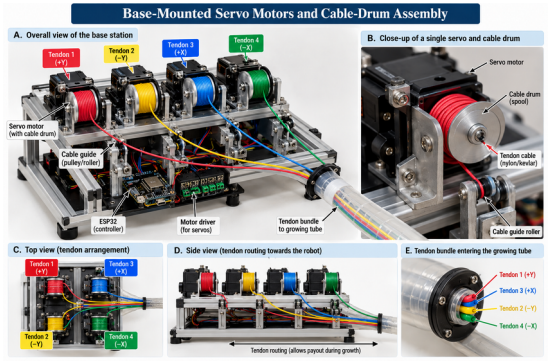


Figure 6.3: Base-Mounted Servo Motors & Cable-Drum Assembly

Tip-Mounted Computer Vision & YOLO Inference Pipeline

From raw camera capture to real-time object classification, bounding boxes, and path planner inputs.

1. CAMERA ACQUISITION & CONVERSION

• Micro-Camera Tip Capture: High-framerate video captures terrain ahead of growing apex.
• ROS 2 Transport & CvBridge: Converts raw sensor buffers into standard OpenCV BGR arrays.

2. PREPROCESSING & LETTERBOXING

• Aspect-Preserving Letterboxing: Resizes frames to 640x640 with symmetric padding.
• Pixel Normalization: Scales RGB values [0, 255] -> [0.0, 1.0] for model tensor batching.

3. YOLO INFERENCE & INTEGRATION

• Pretrained Neural Inference: Multi-scale convolutional feature extraction & class prediction.
• NMS & Trajectory Node: Non-Maximum Suppression sends obstacle coordinates to ROS 2.

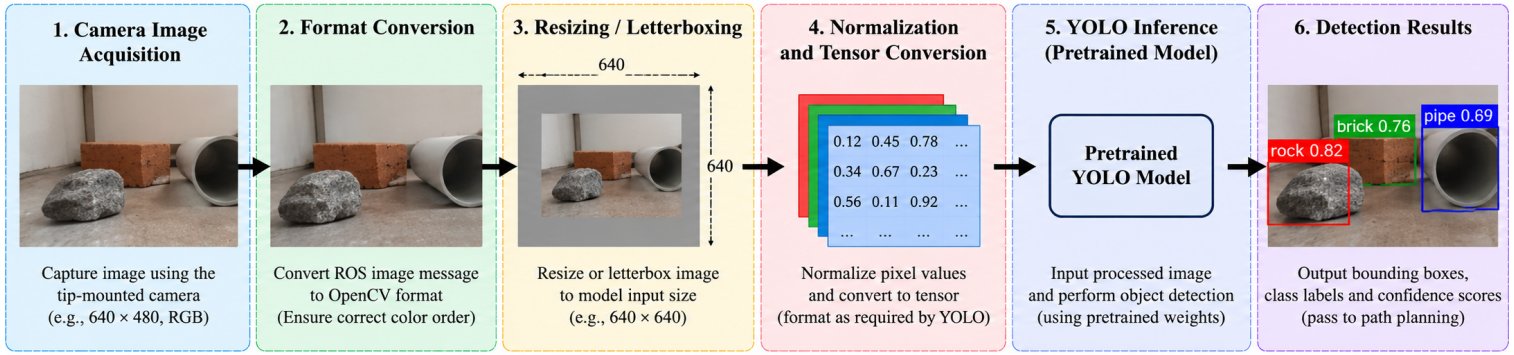


Figure 4.1: End-to-End Image Processing & Pretrained YOLO Object Detection Pipeline

Electronics, Power Distribution & Interfacing

ESP32 embedded microcontroller bridging high-level ROS 2 commands with physical actuators and sensors.

HARDWARE & ELECTRICAL INTEGRATION

- **ESP32 Dual-Core Microcontroller:** Manages low-latency real-time actuator control. Receives high-level trajectory commands from host PC via USB/UART and generates high-resolution PWM signals.
- **Dedicated High-Current Servo Bus:** Steering requires significant torque; four servos are powered by an isolated 5V/6V high-amperage power supply with common ground isolation to prevent logic brownouts.
- **Pneumatic Solenoid Valve Driver:** Transistor/MOSFET switching circuit controls the air flow solenoid valve, modulating pneumatic growth rate and emergency shutoff.
- **Sensory & Safety Interfacing:** Integrated pressure sensor continuously transmits pneumatic line pressure to prevent tube over-inflation; supports optional tip IMU sensor for orientation.

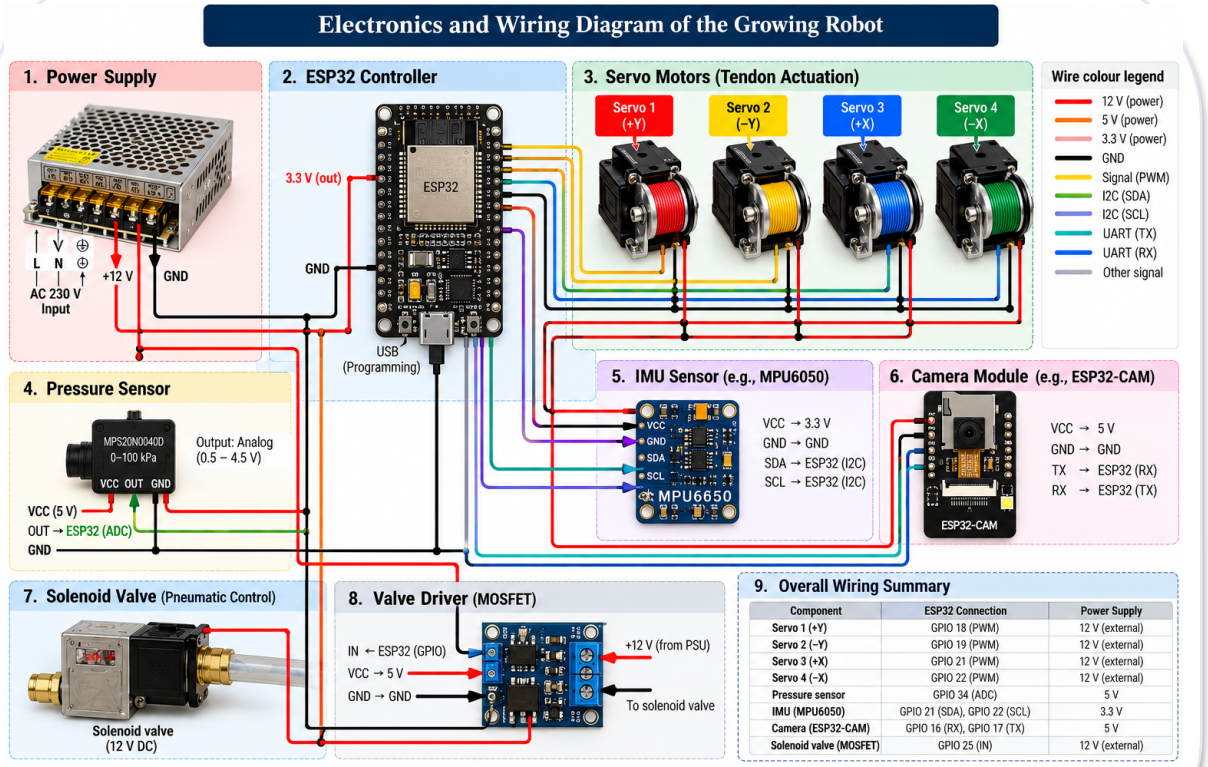


Figure 6.4: Electronics and Wiring Diagram of the Growing Robot

Project Summary, Evaluation Benchmarks & Future Scope

Synthesizing key design achievements and outlining empirical test plans for the prototype.

PROJECT DESIGN ACHIEVEMENTS

- **Integrated Soft-Growing Architecture:** Successfully established a complete design paradigm combining pneumatic tip eversion with 4-tendon steering.
- **Actuator Weight Decoupling:** Positioned high-torque servos at the base station, eliminating moving tip mass and maximizing navigation agility.
- **Vision-Integrated Control:** Formulated an end-to-end perception pipeline leveraging pretrained YOLO for real-time obstacle detection.
- **Modular ROS 2 Architecture:** Created a decoupled framework where mechanical, pneumatic, electronic, and software components can be tested independently.

PLANNED EVALUATION BENCHMARKS

- **Pneumatic Growth Rate:** Measure deployment velocity (cm/s) as a function of air pressure (kPa) and tube material friction.
- **Steering Precision & Deflection:** Quantify maximum angular deflection range (0°–90°), hysteresis, and repeatable steering accuracy.
- **Perception Latency:** Benchmark YOLO frame rate (FPS), detection mAP, and classification confidence under dim lighting.
- **Deployment & Retrieval Cycles:** Test complete forward growth and tendon-assisted retraction without entanglement or membrane rupture.

FUTURE ENHANCEMENTS

- **Closed-Loop Visual Servoing:** Incorporate stereoscopic / depth camera sensing for exact metric 3D obstacle distance measurement.
- **Tension Feedback Control:** Integrate inline miniature load cells on cable drums for closed-loop tension regulation.
- **Granular Media Burrowing:** Evaluate penetration capabilities in noncohesive dry granular media (sand, loose gravel).
- **Autonomous Path Replanning:** Implement dynamic obstacle avoidance and SLAM in GPS-denied confined cavities.